

RESEARCH ARTICLE

Rodents on the Move: Exploring Habitat-Driven Richness, Abundance, and Diversity in East African Woodland

Upendo D. Mollel¹  | Geoffrey E. Soka¹ | Apia W. Massawe^{2,3}

¹Department of Wildlife Management, Sokoine University of Agriculture, Morogoro, Tanzania | ²Institute of Pest Management, Sokoine University of Agriculture, Morogoro, Tanzania | ³African Centre of Excellence for Rodent Pest Management and Biosensor Technology Development, Sokoine University of Agriculture, Morogoro, Tanzania

Correspondence: Upendo D. Mollel (upendaudi@gmail.com)

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ABSTRACT

It is well established that different habitats support different levels of species richness, abundance, and diversity. This is true when it comes to heterogeneous habitats with different habitat conditions. With that focus, we present the findings of our results, which aimed to determine how species richness, abundance, and diversity of rodents vary across different habitats of an East African woodland ecosystem. The study was carried out during the rainy season (January–June 2023), and study grids were marked in four randomly selected habitats, including miombo woodland, *Acacia–Combretum* woodland, riverine, and farmland. Sherman live traps, pitfall traps, and havarhat traps were used to capture rodents in the selected habitats. A total of 219 individuals of rodents were captured in 28,224 trap-nights. These comprised 6 species of rodents, namely *Mastomys natalensis*, *Mus munitoides*, *Acomys wilson*, *Gerbilliscus leucogaster*, *Graphiurus murinus*, and *Paraxerus ochraceus*. The highest abundance of rodents was recorded in farmland and miombo woodland, while the least abundance was recorded in riverine and *Acacia–Combretum* woodland. Species diversity was highest in farmland and miombo woodland and lowest in riverine and *Acacia–Combretum* woodland. The distribution of species varied among habitats. *Acomys wilson* was the most widely distributed species, whereas *Paraxerus ochraceus* was the least. This study suggests that habitat heterogeneity is a key factor influencing rodent richness, abundance, and diversity; hence, it emphasizes the role of wildlife managers and conservationists to protect and conserve these habitats for the survival of small mammal communities.

1 | Introduction

Rodents represent a group of small mammals with diverse distribution across a wide range of habitats, from terrestrial habitats such as natural woodland and farmlands to semi-aquatic habitats including riverine and marshlands (Aplin et al. 2003; Delany 1974; Nel 1978; Nowak and Walker 1999). Rodents play critical ecological, social, and economic roles (Tadesse and Bekele 2008), which contribute to the ecological pyramid where they take part in a food web in an ecosystem, serving as a food source to different predators, hence supporting diversity for the

sustenance of carnivores and raptors (Hanski et al. 1991; Tadesse and Bekele 2008; Torre et al. 2007). They serve as valuable indicators of ecological integrity and in predicting environmental changes (Avenant and Cavallini 2007; Shilereyo et al. 2020). The presence of rodents in various habitats has led scientists to examine their spatial, temporal, feeding, behavioural, and movement patterns. These studies have consistently demonstrated that the structure of rodent diversity is shaped by factors such as vegetation structure, food availability, and the presence of predators, which are crucial for their continued survival and existence across habitats (Mayeji et al. 2025; Mulungu et al. 2008,

2011; Odhiambo et al. 2008; Mwasapi and Rija 2022; Layme et al. 2004).

The role of habitat type, vegetation structure, and composition in shaping rodent diversity and community assemblages is explained by the habitat heterogeneity hypothesis (Stevens and Tello 2011; Thomas et al. 2022). Originally proposed by MacArthur and MacArthur (1961), this hypothesis posits that heterogeneous habitats sustain higher species richness and diversity because an increase in microhabitats provides more ecological niches that facilitate species coexistence (MacArthur and MacArthur 1961; August 1983; Stein and Kreft 2015; Mayamba et al. 2019). Consequently, rodent species richness, abundance, and diversity are strongly influenced by habitat characteristics, particularly vegetation type and cover (Carrilho et al. 2017; Chidodo et al. 2020; Ssuuna et al. 2020). Previous studies have demonstrated the role of habitat heterogeneity and fragmentation in shaping rodent diversity. However, questions remain on how woodland-dominated heterogeneous habitats in protected areas influence rodent abundance, richness, and diversity. The Wami-Mbiki Game Reserve, a transition zone from coastal mangroves to deciduous woodlands and lowland forests, supports diverse wildlife. Initially designated as a Wildlife Management Area in 2002, it was upgraded to a Game Reserve in 2021 to strengthen habitat and biodiversity conservation, a shift that may enhance both plant and animal communities. Most studies in East African protected ecosystems, including the Wami-Mbiki Game Reserve, have primarily focused on large mammal species due to their importance in wildlife tourism

activities (Bukombe et al. 2022). In contrast, there is limited information on the abundance, species richness, and diversity of small mammals, particularly rodents, within these ecosystems. Assessing rodent communities is important for achieving a comprehensive biodiversity inventory, which is essential for implementing effective and sustainable wildlife and ecosystem conservation. This study aimed to evaluate how different habitats influence rodent richness, abundance, and diversity, using species documentation as a means to understand their diversity patterns across habitats. We hypothesize that different habitats significantly influence rodent community composition, leading to variation in species richness, abundance, and diversity. In line with our hypothesis, the study addresses the following key ecological questions: First, how does the abundance of rodents differ across habitats? Second, what is the difference in rodent species richness and diversity inside the reserve (natural habitat) and outside the reserve (modified habitat)?

2 | Materials and Methods

2.1 | Study Area Description

The study was conducted in Wami-Mbiki Game Reserve, which is located in the central eastern part of Tanzania, situated in the Morogoro and coastal regions of Tanzania. Wami Mbiki Game Reserve is located approximately 100 km off the Indian Ocean coast. It lies between latitudes 06°10' and 06°30' S and longitudes 37° and 38° E (Figure 1). The reserve was

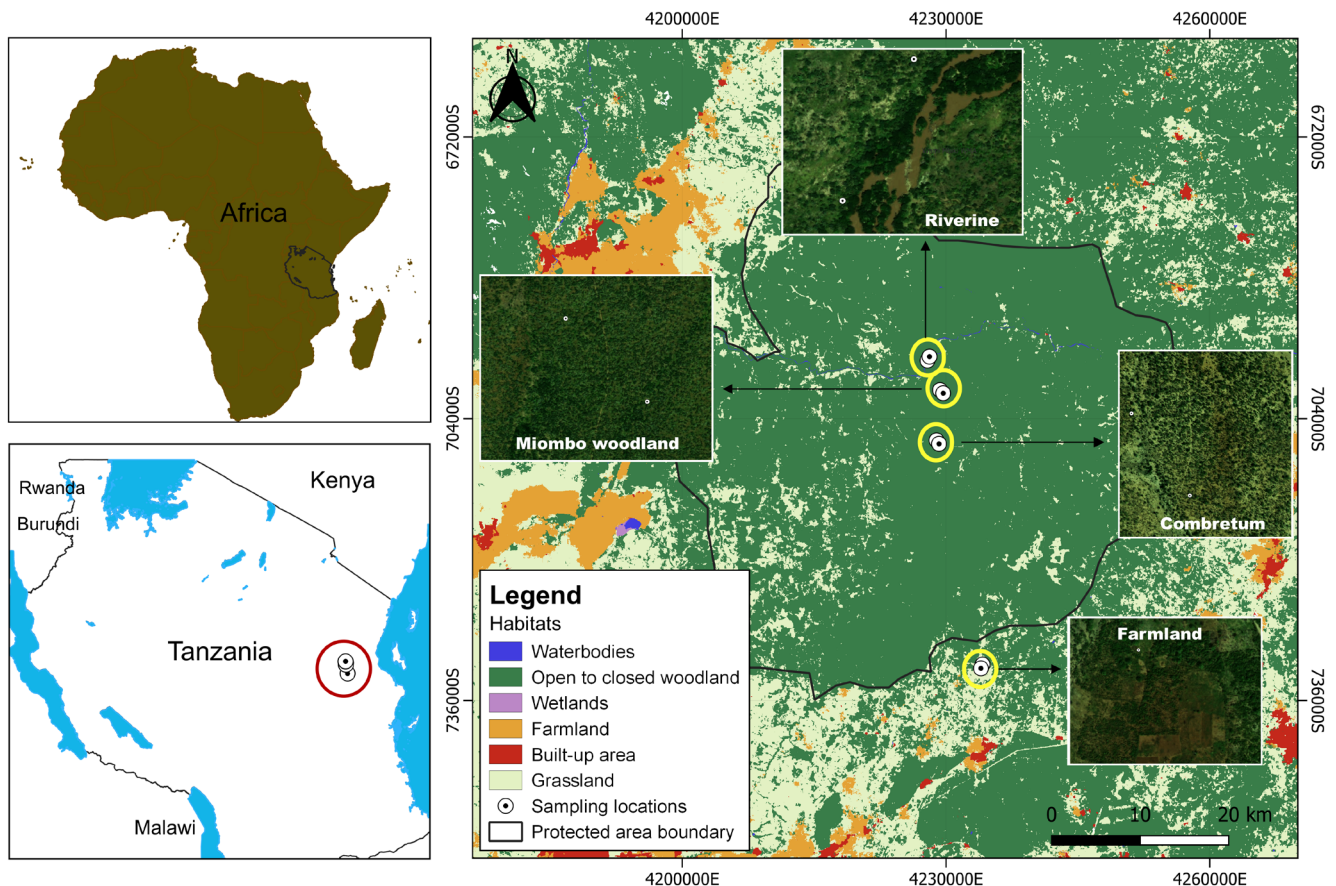


FIGURE 1 | The land use map of the study area showing different habitats and rodent sampling locations.

previously known as the Wami-Mbiki Wildlife Management Area, which was then upgraded to Game Reserve, covering an area of 2466.52 km². The reserve falls within the warm tropical climate, which experiences short rains in October, November, and December, followed by long rains from March to May. The average annual rainfall ranges between 700 mm and 900 mm, and the average temperature ranges between 26°C and 28°C. This ecosystem is inundated with clay, sand, and loam soils and is covered with different types of vegetation communities, mainly the Miombo woodlands, which cover about 89%, dominated by *Julbernardia globiflora* associated with *Brachystegia speciformis*; *Acacia-Combretum* woodland covers about 8.1%, dominated by acacia species which are *A. nigrescens* and *A. robusta* associated with *Sclerocarya birrea*, *Terminalia stuhlmanii*, and *T. stenostachya* tree species; Riverine or riparian cover about 1.1%, which is dominated by *Albizia petersiana*, *Terminalia spinosa*, and *Combretum imbebe* associated with palm *Hyphaene* and Phoenix; and patches of thickets dominated by the *Grewia*, *Dichrostachys*, and *Acalypha* with succulent *Euphorbia*. The reserve forms part of important wildlife corridors connecting Saadani, Mikumi, Nyerere National Parks, and Selous Game Reserve. Different animals such as greater kudu, wildebeest, elephant, buffalo, zebra, impala, waterbuck, duiker, hyena, lion, wild dog, crocodiles, yellow baboon, velvet monkeys, bush pig, and hippo are found. Additionally, the reserve hosts a famous river, the Wami River, which harbors different fish species, including tiger fish and the giant catfish.

2.2 | Study Design

The study was purposely conducted during the rainy season between January 2023 and June 2023 in four different habitat types, including miombo woodland, *Acacia-Combretum* woodland, riverine, and farmland. Two methods—Capture-Mark-Recapture and pitfall method—were employed for rodent trapping. CMR used permanent trapping grids of 70×70 m, where each 70×70 m grid consisted of 7 parallel lines spaced 10 m apart, each with 7 trap stations. Therefore, each grid consisted of 49 traps (Makundi et al. 2010; Seviyanu and Stermin 2015), and a minimum distance of 500 m was established from one grid to another within the same habitat. A mixture of peanut butter and maize flour was used as bait placed in Sherman live traps measuring 23 × 9.5 × 8 cm, following Mulungu et al. (2008). Four (4) harvath traps (60×15×170 cm) were also established randomly at each habitat for trapping larger species (Welegerima et al. 2020) and baited with bananas and carrots. All the captured individuals in CMR were distinctly marked through toe clipping, a method considered ecologically acceptable and commonly applied in tropical studies (Leirs et al. 2023, 1997; Massawe et al. 2011; Mayeji et al. 2025). This marking technique facilitated reliable identification of recaptured individuals during subsequent counts. In this study, we used the approach of recording species abundance and calculating diversity across habitats, which is a common approach in ecological studies. Although the sampling period did not encompass the complete seasonal dynamics, it was adequate for examining spatial variation in rodent abundance, richness, and diversity among habitats, as supported by Leirs (1995); Leirs et al. (1997); Massawe et al. (2011); Mayeji et al. (2025); Smith (1968). The main limitation was the restricted timeframe; however, this was

mitigated by conducting repeated sampling each month over five consecutive nights (Massawe et al. 2011; Mayeji et al. 2025).

In pitfall method, one drift-fence was established randomly along pitfall lines, each comprising 11 buckets, placed 5 m apart, and buried in the ground so that the top of the bucket levels with the ground (Bayo 2019; Shilireyo et al. 2020). Each pitfall line had a 50 cm high black plastic drift fence running over the center of each bucket for leading the rodents into the traps. All traps were left for three consecutive nights, and information such as species name, sex, weight, and number of species trapped was recorded for further use and identification purposes. The taxonomic nomenclature of all captured rodents was identified to the species level following Kingdon (1997) and Happold (2013).

2.3 | Data Analysis

Data were gathered in excel version 2021 prior to analysis, the abundance data which was the primary collected data were gathered for analysis for each habitat. The species richness and diversity across habitats were calculated using the Chao1 estimate and Shannon diversity index respectively under the software EstimateS (Colwell 2013). Species richness, Shannon Wiener, evenness, and dominance were then estimated from the computed indices. Species richness was obtained as the number of different species in each habitat type, whereas rodents' abundance was calculated as total counts of captured individuals per each species recorded. Species diversity was estimated using standard Shannon wiener index; $H' = -\sum_{i=1}^s (P_i * \ln P_i)$. Where H' represents content of sample, index of species diversity or Degree of Uncertainty, s represents the total number of species P_i represents the propositional of total sample belonging to i^{th} species (Mulungu et al. 2008; Shannon and Weaver 1949). Kruskal Wallis test was used to compare the species composition, richness and diversity data between habitat types following a non-normal distribution confirmed using Shapiro Wilk test (Shapiro test, $p > 0.05$). Species indicator analysis was performed in R version 4.4.2 using 'ggplot2' package. Additionally, the PAST Paleontological Statistics software V4.13 (Hammer et al. 2002) was used to perform correspondence analysis (CA) at the correlation coefficient ($r \geq 0.5$). For the analysis, only data collected through the CMR method were used. Although pitfall traps were deployed, they did not capture any rodent species but instead trapped only invertebrates hence, they were excluded from the analysis dataset.

3 | Results

3.1 | Rodent Species Composition Across Different Habitats

A total number of 219 individuals from 6 species of rodents were trapped on 28,224 trap nights, of which 26 individuals were recaptured. *Acomys wilson* and *Mastomys natalensis* were the most abundant species, which contributed 79.45% of the total captures. *Acomys wilson* was captured in all habitats; hence, it dominated both habitat types with 47.49% (Table 1). *Mastomys natalensis* dominated the farmland (54.92%), while other species including *Mus munitoides*, *Graphiurus murinus*, *Gerbilliscus leucogaster*,

and *Paraxerus ochraceus* had the lowest percentage composition of the total captures, with 7.31%, 5.02%, 4.11%, and 4.11%, respectively. Interestingly, the majority of the species occurred across habitats (Table 1; Figure 2); however, the Chi-square test indicated that the percentage composition (occurrence) of sampled species varied significantly across habitats ($\chi^2=377.72$, $df=5$, $p<0.001$). For example, *Acomys wilson* ($\chi^2=11.09$, $df=3$, $p=0.011$), *Graphiurus murinus*, *Mastomys natalensis*, *Mus munitoides*, and *Paraxerus ochraceus* ($\chi^2=8.32$, $df=3$, $p=0.03$). The percentage composition of *Gerbilliscus leucogaster* did not differ significantly across habitats ($\chi^2=4.5$, $df=3$, $p=0.21$).

In terms of temporal variations in abundance, there was no significant difference between sampled months ($\chi^2=6.33$, $df=5$, $p=0.27$). Overall, the abundance was higher in June and lower in April (Figure 3); however, in farmland, abundance was high in February and lower in May (Figure 4). A relatively higher mean abundance ($\pm$ SE) was recorded in the months of June (16 ± 5.1), February (9.8 ± 6.8), and May (9.8 ± 3.1). More species were observed in February, May, and June (Figure 2). Fewer species were observed in April, which differed from the rest, while in January in *Acacia-Combretum* and miombo woodlands, no rodent species were captured (Figure 4).

TABLE 1 | Rodent species composition represented in percentages (number in parentheses) across habitats. The assigned abbreviations correspond to different habitat types.

Species	Habitat				Total
	ACW	FL	MWL	RVR	
<i>Acomys wilson</i>	17 (85)	39 (31.97)	26 (57.78)	22 (68.75)	104 (47.49)
<i>Gerbilliscus leucogaster</i>	0 (0)	9 (7.38)	0 (0)	0 (0)	9 (4.11)
<i>Graphiurus murinus</i>	0 (0)	0 (0)	7 (15.56)	4 (12.50)	11 (5.02)
<i>Mastomys natalensis</i>	3 (15)	67 (54.92)	0 (0)	0 (0)	70 (31.96)
<i>Mus munitoides</i>	0 (0)	0 (0)	10 (22.22)	6 (18.75)	16 (7.31)
<i>Paraxerus ochraceus</i>	0 (0)	7 (5.74)	2 (4.44)	0 (0)	9 (4.11)
Total	20 (100)	122 (100)	45 (100)	32 (100)	219 (100)

Abbreviations: ACW, *Acacia-Combretum* Woodland; FL, Farmland; MWL, Miombo Woodland; RVR, Riverine. The codes correspond to abbreviations of habitat types.

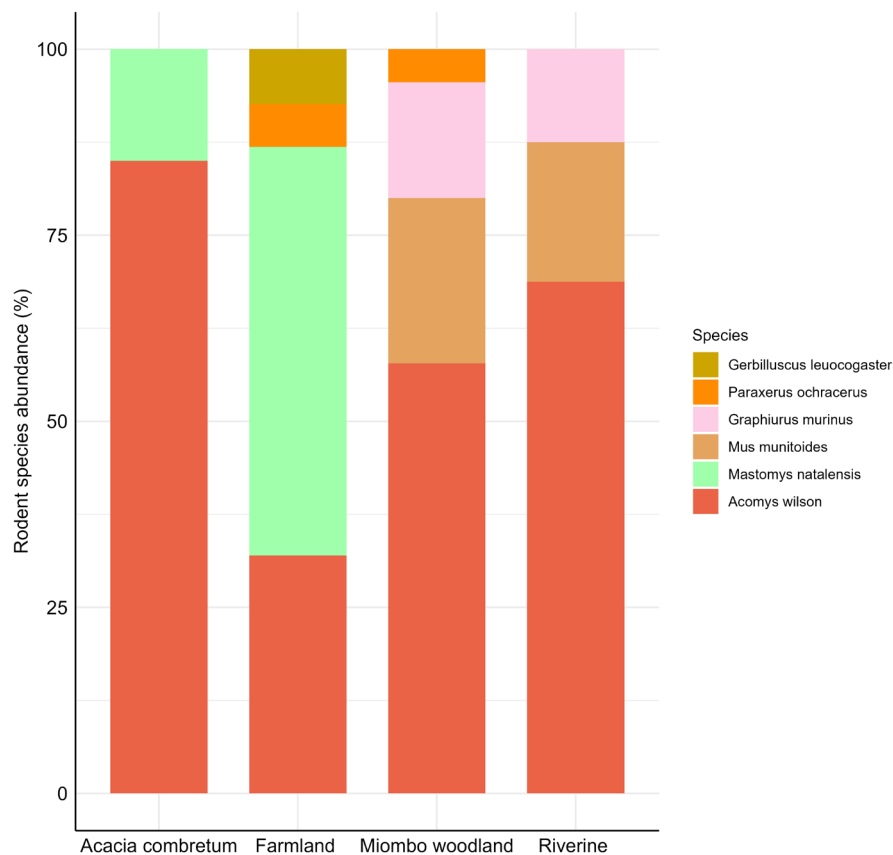


FIGURE 2 | Species composition of rodents based on relative abundance across four different habitats.

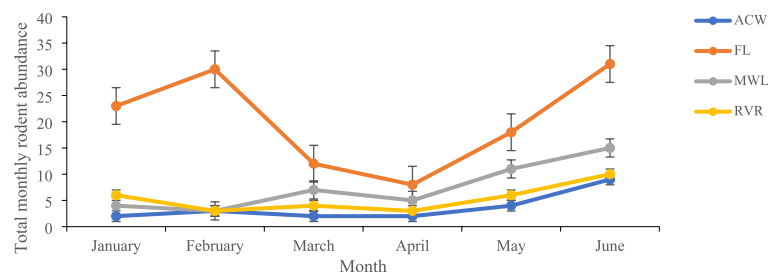


FIGURE 3 | Mean ($\pm$ SE) monthly rodent abundance across the four habitats over the study period. ACW, *Acacia-Combretum* Woodland; FL, Farmland; MWL, Miombo Woodland; RVR, Riverine.

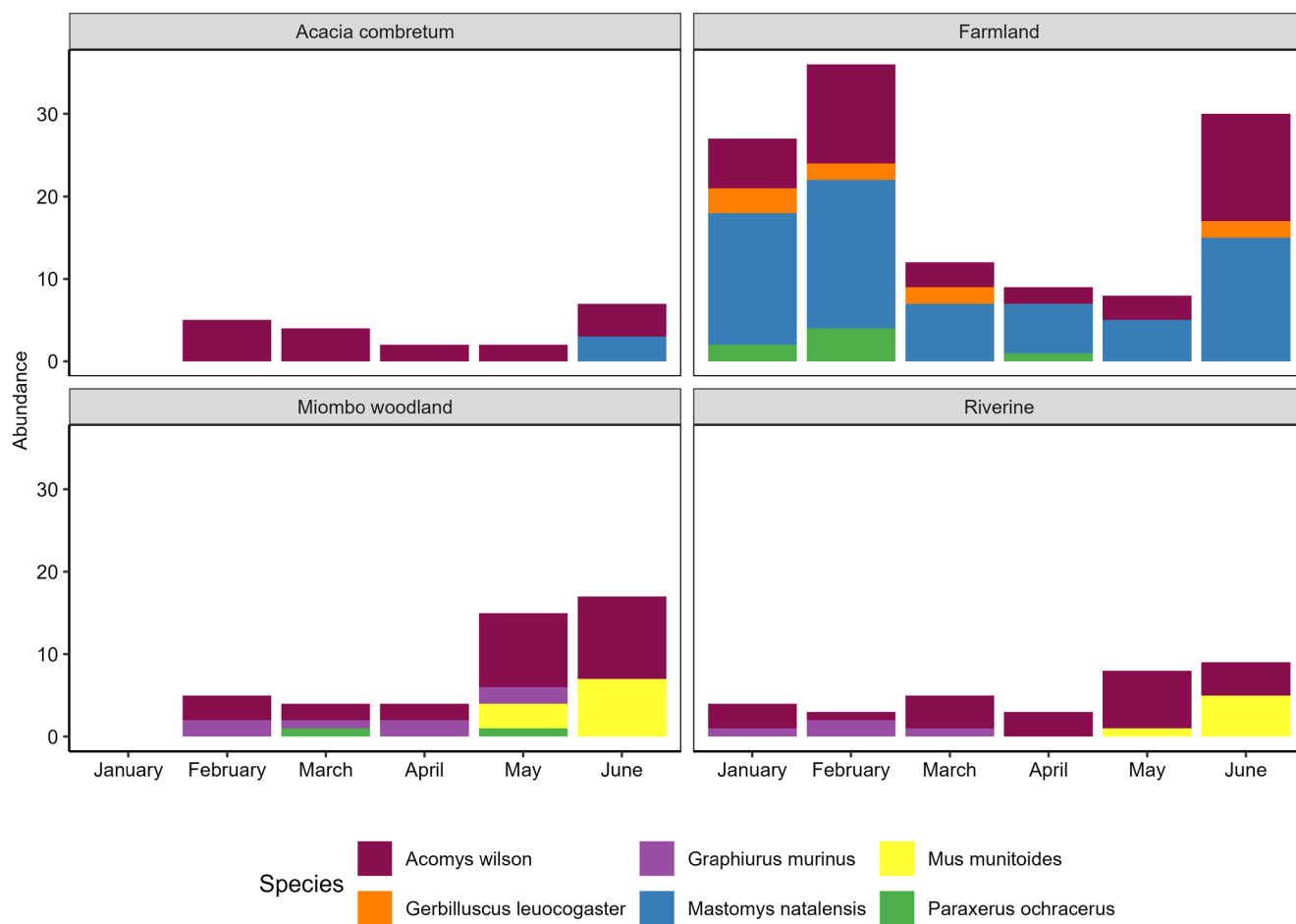


FIGURE 4 | Monthly abundance and composition of rodent species across habitats.

In addition, correspondence analysis explained the variations in two axes (Figure 5). Axis 1 explained 80.19% of the association. *Paraxerus ochraceus*, *Gerbilliscus leucogaster*, and *Mastomys natalensis* associated with farmland. Further, *Mus munitoides* and *Graphiurus murinus* were positively correlated with Miombo woodland and riverine habitats, indicating that *Mus munitoides* and *Graphiurus murinus* were more associated with Miombo woodland and riverine habitat, while *Acomys wilson* was more associated with *Acacia-Combretum* woodland habitat. Furthermore, the indicator species analysis revealed a significant difference ($p < 0.05$) in the distribution and abundance of rodent species across habitats (Figure 6). This variation was particularly evident in species such as *Acomys wilson*, which showed high relative abundance across habitats, indicating its

dominance in all habitats. On the other hand, species such as *Mastomys natalensis*, *Graphiurus murinus*, and *Mus munitoides* showed varying abundance across habitats, with some habitats being higher than others.

3.2 | Rodent Species Richness and Diversity Across Habitats

Six rodent species were captured across different habitats, mainly miombo woodland, *Acacia-Combretum* woodland, riverine, and farmland. As mentioned before, the species captured were *Acomys wilson*, *Gerbilliscus leucogaster*, *Graphiurus murinus*, *Mastomys natalensis*, *Mus munitoides*, and *Paraxerus*

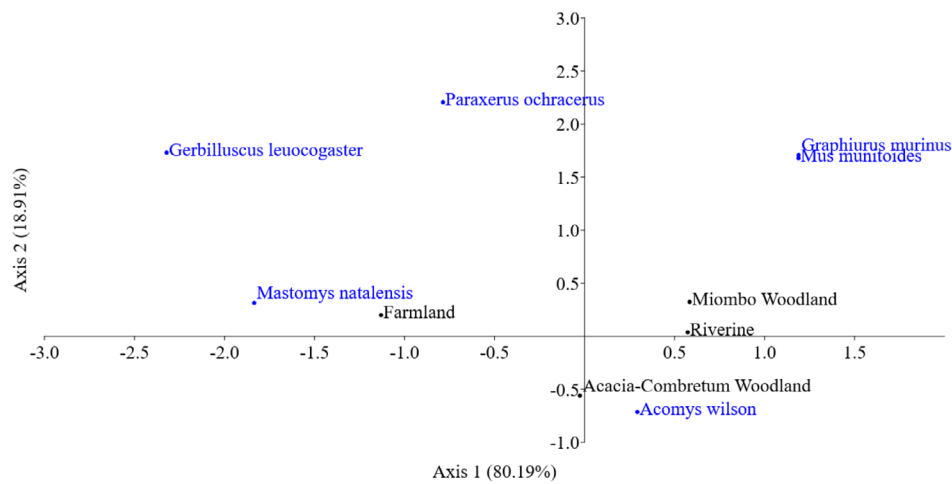


FIGURE 5 | Ordination plots showing rodent species associated with different habitats in Wami Mbiki Game Reserve. Axis 1 and 2 explained 80.19% and 18.91% of the variation in rodent association, respectively.

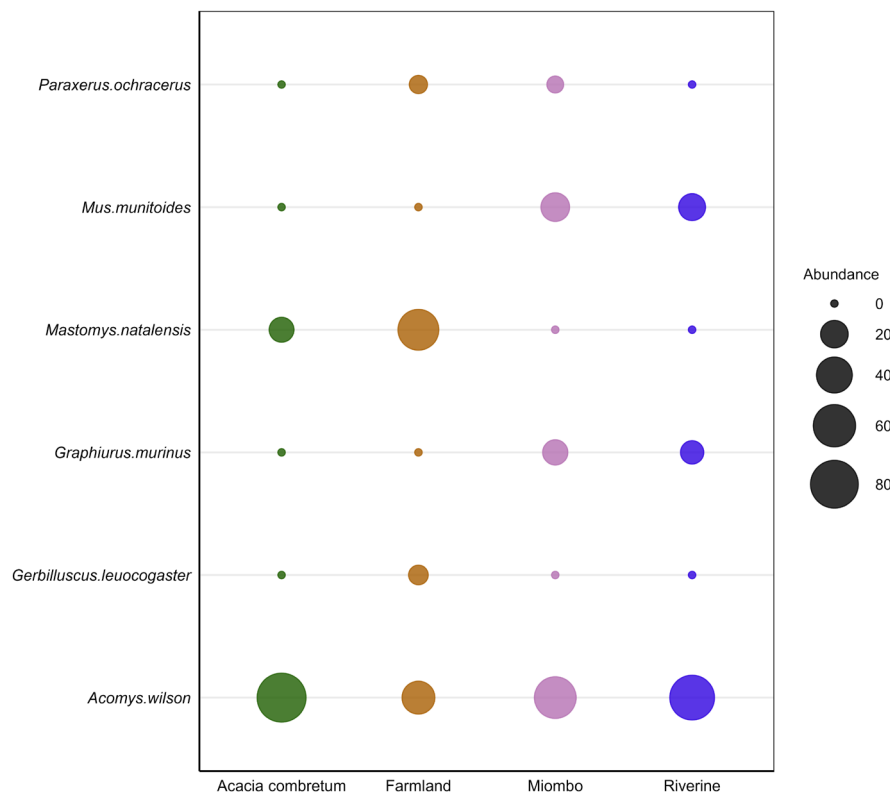


FIGURE 6 | Bubble plot showing dominance of rodent species in terms of abundance across habitats.

ochraceus (Table 1). The rodent species richness was highest in farmland and miombo woodland ($n=4$), followed by riverine ($n=3$), and lowest in *Acacia-Combretum* woodland ($n=2$). Rarefaction curves showed the climax saturation of trapping effort for all habitats, indicating a good sampling effort with minimal encounters of new species after several captures (Figure 7). Rodent species diversity was highest in farmland, followed closely by miombo woodland habitat, and lowest in *Acacia-Combretum* woodland (Figure 8). Species evenness was high in the miombo woodland, followed closely by riverine and farmland (Figure 8). There was no significant difference in species richness (Kruskal-Wallis test: $\alpha = 0.05$, $H' = 4.23$, $df = 3$ and

$p = 0.542$) and diversity (Kruskal-Wallis test: $\alpha = 0.05$, $H' = 2.57$, $df = 3$ and $p = 0.4173$) between habitat types.

4 | Discussion

4.1 | Rodent Composition Across Habitats

Heterogeneous habitats in most ecosystems have been associated with high species richness and diversity (Kavana et al. 2024; Lyimo et al. 2024), which supports the hypothesis that heterogeneous habitats support high species richness;

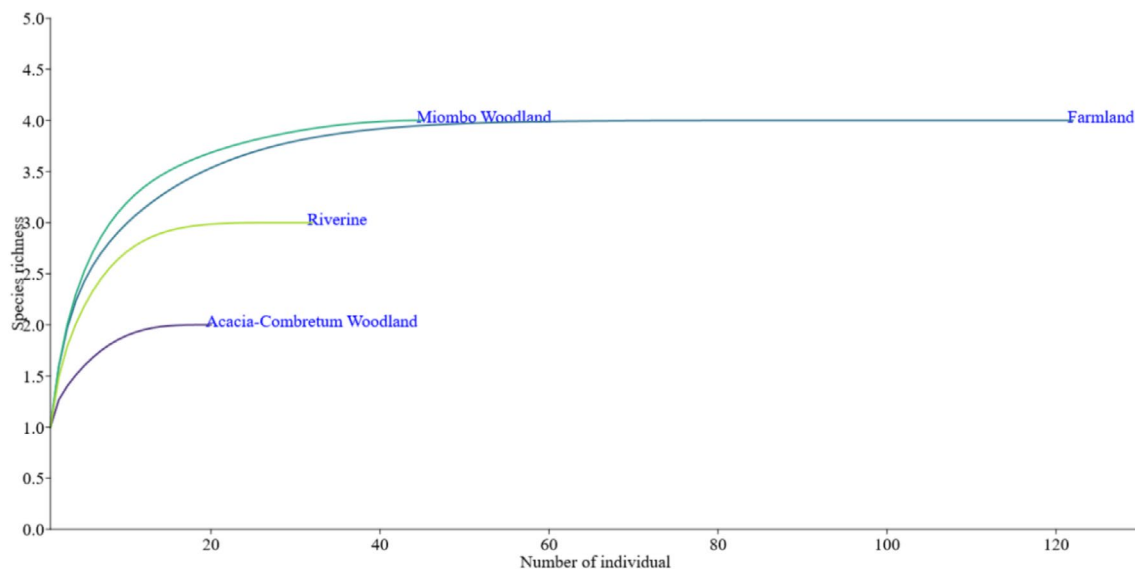


FIGURE 7 | Rarefaction curves of species richness plotted against cumulative number of individuals.

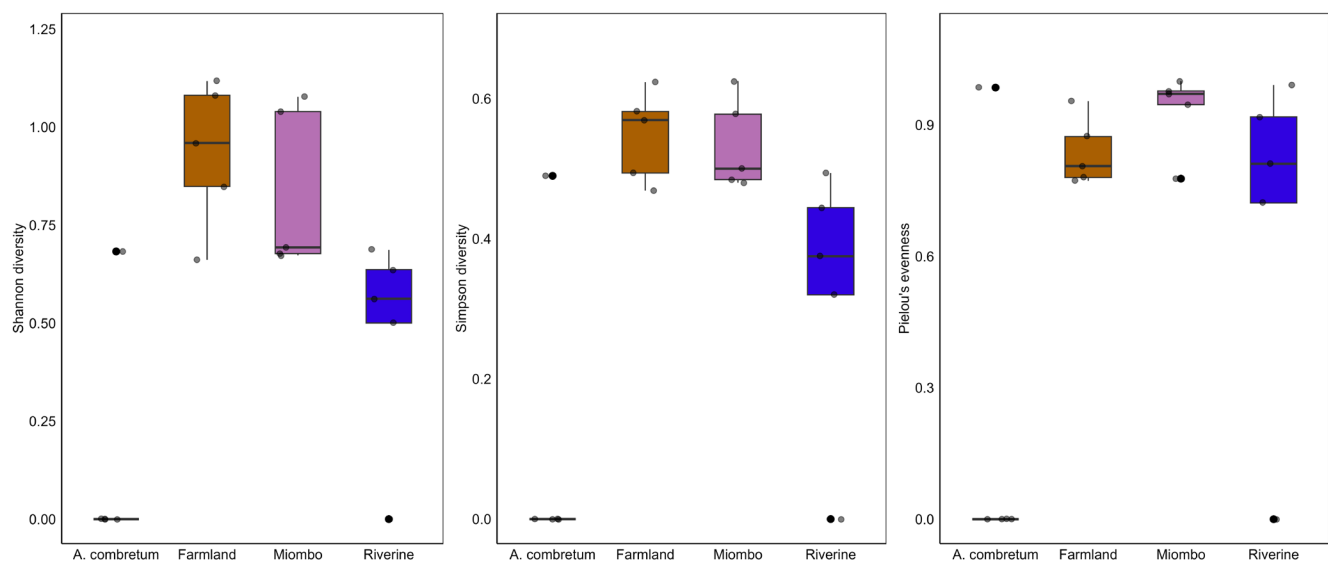


FIGURE 8 | Rodent species diversity and evenness based on Shannon, Simpson, and Pielou indices. The p values on top of box plots indicate significance at $p < 0.05$.

hence, high species diversity, as microhabitats contribute to the availability of more niches, which favor species coexistence (MacArthur and MacArthur 1961). In our study, we report the abundance, richness, and diversity of rodent species across heterogeneous habitats of the Wami-Mbiki Game Reserve. Farmland, which stands as human-modified habitats, exhibited high species richness and hence high species diversity, indicating the role of human-modified habitats in supporting species diversity (Mayeji et al. 2025). High species richness and diversity in farmland may be influenced by factors such as food availability and vegetation cover. In contrast, natural habitats like miombo woodland, riverine, and *Acacia-Combretum* woodland supported fewer species, possibly due to their complex structure that favors predators and limits rodent diversity.

Among the six species trapped, *Acomys wilson* was present in all habitats. This opportunistic species thrives in dense

vegetation, colonises various habitats, and responds readily to environmental changes, making it a generalist (Fleming and Nicolson 2002; Massawe et al. 2005). *Gerbilliscus leucogaster* and *Paraxerus ochraceus* were the least abundant, found in farmland and miombo woodland, respectively. The low abundance of *G. leucogaster* in farmland may be due to limited food, as much of the land was under preparation for cultivation, although this species is generally widespread and able to persist in diverse habitats (Monadjem et al. 2015; du Plessis et al. 2016). Our results support previous studies by indicating that *A. wilson* and *M. natalensis* are among the most adaptable and widespread rodents in East Africa (Massawe et al. 2005). *M. natalensis* was the second most captured, occurring primarily in farmland and *Acacia-Combretum* woodland. Its high abundance in farmland indicates its adaptability to both natural and human-modified habitats. This generalist species has a wide distribution and is commonly associated

with agricultural fields and human dwellings, but also inhabits natural savannas and grasslands, often showing population surges following disturbances (Makundi et al. 2007; Mulungu et al. 2013; Monadjem et al. 2015). *G. murinus* was primarily found in miombo woodlands and riverine forests, utilising tree holes and rock crevices for shelter, as reported by Madikiza (2010); Lamani (2011) and Madikiza (2017). In our study, most individuals were captured in miombo woodland rather than in riverine habitats, suggesting a preference for woodlands that offer abundant resting sites. Except for *Acacia-Combretum* woodland, the four other habitats showed distinct rodent communities; however, miombo woodland and riverine habitats shared similarities, likely due to their adjacency and comparable vegetation, which may attract similar species.

Additionally, *P. ochraceus*, *G. leucogaster*, and *M. natalensis* were found exclusively in farmland habitats, while *G. murinus* and *M. minutoides* were mainly associated with miombo woodland and riverine habitats. This pattern suggests that rodent species differ in their habitat preferences, with some being specialists and others generalists. *G. murinus* and *M. minutoides* appear to favour dense vegetation, consistent with previous studies indicating that species like *G. murinus* are generalists capable of occupying a wide range of habitats (Kingdon 2003; Fitzherbert et al. 2006; Skinner and Chimimba 2005; Madikiza 2010; Lamani 2011). These species primarily nest in woodlands but can also use rocks, underground burrows, tree branches, shrubs, abandoned bird nests, and beehives. In this study, the observed variation in rodent distribution among habitats is linked to habitat characteristics. Farmland, miombo woodland, and riverine habitats provide sufficient vegetation cover, offering shelter and food resources that support higher rodent captures, whereas *Acacia-Combretum* woodland, with its open canopy and sparse understory, supports fewer individuals.

4.2 | Species Richness and Diversity Across Habitats

Farmland and miombo woodland have been shown to support higher rodent species richness compared to *Acacia-Combretum* woodland and riverine habitats, with a similar pattern observed for species diversity across the four surveyed sites. Higher species richness in farmland and miombo woodland likely reflects more favorable conditions for rodents, including diverse plant communities that provide abundant food resources, as evidenced by our results. Previous studies indicate that rodent species richness is strongly influenced by habitat productivity, with resource-rich areas supporting higher numbers of species (Mayamba et al. 2020). Variations in habitat preference may be shaped not only by food availability but also by food quality (Lobo and Millar 2011). Habitat heterogeneity in farmland and miombo woodland may also offer better shelter for rodents compared to riverine and *Acacia-Combretum* woodland habitats. Higher species richness in farmlands may be related to the intermediate disturbance hypothesis (IDH) as introduced by Connell (1979), which suggests that minimum levels of disturbance can promote species survival through preventing competitive exclusion, hence allowing multiple species to coexist. The

high diversity in miombo woodland could be attributed to the fact that such habitats have a wide diversity of plants, which may translate into higher food availability, attracting rodents and resulting in increased rodent diversity. This finding is consistent with the findings from previous studies (Leis et al. 2008; Ssuuna et al. 2020).

Farmland has been reported to be the second most diverse habitat in this study, with its high population being attributed to abundant food and a lower abundance of predators when compared to natural forest habitats. In the present study, we attribute the *Acacia-Combretum* woodland's low diversity (low species richness and abundance) to its open canopy and lack of understory vegetation. Such variations could also be explained by human activity, which modifies habitat, hence altering rodent communities (Hoffmann and Zeller 2005). Therefore, our findings point out that the difference in species composition reflects the influence of habitat as key factors for the abundance and diversity of rodents, where they are not only applicable in East African woodland but to ecosystems worldwide. A similar pattern has been observed in other regions such as the tropical region and temperate region, as mentioned by Poindexter et al. (2012) and Cramer and Willig (2002), implying that habitat heterogeneity is a major driver of rodent population dynamics worldwide. Understanding their abundance and diversity across different ecosystems helps us predict ecological responses to environmental changes and provides important insight for the conservation of biodiversity, pest management, and ecosystem health on a regional and global scale.

5 | Conclusion

Based on the evidence of our findings, there is observable variation in rodent species richness, abundance and diversity across the study habitats in Wami Mbiki Game Reserve. Farmland exhibited the highest composition, followed by miombo woodland, while riverine and *Acacia-Combretum* woodlands had the lowest values. Farmland supports a higher abundance of rodents due to plenty of food supplies and microhabitats. These findings emphasise the importance of habitat features in shaping rodent population.

The implication of these findings suggests that the low diversity in natural habitats might indicate habitat degradation and there is a need to improve conservation strategies. However, the diversity of rodent species in farmland and miombo woodland indicates that these habitats provide good shelter and food sources for rodents. While the scope of this study focused on habitat-driven species richness, abundance, and diversity of rodents in East African woodland, future research could explore factors such as seasonality variation and breeding season on rodent populations. With these findings, we therefore advocate the role of conservation managers and environmental planners in facilitating the conservation of rodents and the entire community of small mammals through conservation approaches that aim at conserving other biodiversity. Conservation efforts should span within and outside protected areas through maintaining habitat connectivity and adaptive conservation approaches for biodiversity conservation.

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Ethics Statement

Ethical approval was granted by the Tanzania Commission for Science and Technology (COSTECH) in collaboration with the Tanzania Wildlife Research Institute (TAWIRI) permit no: 2023-051-NA-2023-1469. An entry permit into the Wami Mbiki Game Reserve was granted by the Tanzania Wildlife Management Authority (TAWA).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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